FLINDERS UNIVERSITY

College of Science and Engineering

Major: Information of Technology

REPORT OF ASSIGNMENT

CLOUD AND DISTRIBUTED COMPUTING GE

COMP9033_2025_S1_FP

Topic coordinator: Dr. Alireza Jolfaei

Student name: Le Thanh Nguyen

Student FAN: nguy1710

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Dr. Alireza Jolfaei**, my lecturer, who provided me with a fundamental and in-depth knowledge of cloud computing, enabling me to carry out this project systematically and effectively.

I would also like to thank to **Southern Power Corporation (Vietnam)** – my previous company - for allowing me to use Azure Cloud computing resources from company's projects I was still undertaking to support building this academic assignment. This support played a crucial role in helping me to have budget to deploy the necessary cloud services for the project.

Special thanks to the **Adelaide Metro Developer team** for providing public real-time API access, which made it easier for me to retrieve actual transportation data to build and test the system.

Lastly, I would like to thank my **project team members** for their dedication, collaboration, and teamwork throughout the project. Their effort and commitment were key to completing this work on time and to a high standard.

ABSTRACT

We did this academic project assuming that we are a technology startup named **Orange company** in Adelaide, South Australia. The project's aim is to design an integrated solution based on cloud computing to collect, process, store and analyse real-time data from the Adelaide Metro public transport system.

The system uses Microsoft Azure cloud services, combined with Node-RED platform to automate data processing and alerting. Data collected from Adelaide Metro's public API, stored, and then used to create analytical dashboards on the Tableau platform, supporting management in strategic decision making. Ordinary users can also look up vehicle information through the website (https://adelaidemetrorealtime.azurewebsites.net).

The project not only helps students apply academic knowledge into practice but also hopes to be developed into a useful application, helping people in Adelaide to look up information on public transport more conveniently, in addition to the current Adelaide Metro system.

Table of Contents

Table of Figures

Table of Tables

INTRODUCTION

(Assumption) **Orange Company** is a startup company headquartered in Tonsley, Adelaide, South Australia. The company provides software solutions and IT consulting services in the areas of data analytics, autonomous systems, and artificial intelligence. Orange has a wide range of customers such as companies and organizations.

The "**Adelaide Metro Mining Project**" campaign was established at the request of the client, **Adelaide Metro** - the management unit of the public transport system in Adelaide, under the South Australian Department of Infrastructure and Transport.

The project aims to build a pipeline system on the **Microsoft Azure** cloud computing, combined with **Node-RED** platform, to collect real-time vehicle location data from Adelaide Metro, serving the analysis, monitoring, and optimization of public transport operations in the area.

Scope of Work:

- **Design overall system architecture** to collect and process data from Adelaide Metro
- **Design automation method** to validate, clean and simplify raw data from client's API (Adelaide Metro API)
- **Design database** to store the data collected after cleaning process.
- **Create a web interface** for public users to look up vehicle route information, current vehicle location, travel history, and statistics (average speed, congestion areas, etc.).
- **Develop a warning system** for vehicles traveling below the speed limit, automatically sending emails to Adelaide Metro's managers.
- **Build visualisation** to analyse historical traffic activity, supporting Adelaide Metro to evaluate performance.
- **Use Node-RED as middleware** to operate data collection, alerting, and reporting.

Significance of Development:

The Adelaide Metro Mining Project is expected to support Adelaide Metro in solving current public transport problems as well as planning for infrastructure development in future. Specifically, the project brings the following values:

- Improve the ability to monitor and coordinate traffic operations for Adelaide Metro management through a visualisation system (dashboard) and real-time warnings for vehicles.
- Increase transparency and access to information for users through a simple web interface.
- Contribute to the modernisation of public transport infrastructure management processes, while opening the direction for the development of applications of big data analysis, artificial intelligence and cloud computing technologies in Adelaide.

RELATED WORK

The application of real-time data to the management and optimization of transport systems has been widely deployed around the world. Below are some typical works and projects:

- Transport for NSW (New South Wales, Australia): They adopted a Multi-Modal Performance Reporting Program, using the Microsoft Azure platform to collect and analyse data from transport vehicles such as buses, ferries, light rail and metro. Data is updated every 10 seconds, allowing the creation of dashboards using Power BI to monitor operational performance and improve customer service (Microsoft Corporation, 2020).
- Cubic, a company from California, USA provides real-time passenger information solutions, including vehicle location prediction, alerting passengers to service changes, and integration with station information display systems (Cubic Corporation, 2021).

BODY OF REPORT

Requirements and Constraints

Requirements

1. **Get real-time data from Adelaide Metro API** (https://gtfs.adelaidemetro.com.au)

The Adelaide Metro API provides real-time Json data of vehicles in the whole Adelaide city. Data will be reloading every 15 seconds. Information such as trip id, vehicle id, route, Start date, location (latitude, longitude), speed, time stamp, airconditioned and disability. The target address API for Orange's project: https://gtfs.adelaidemetro.com.au/v1/realtime/vehicle_positions/debug

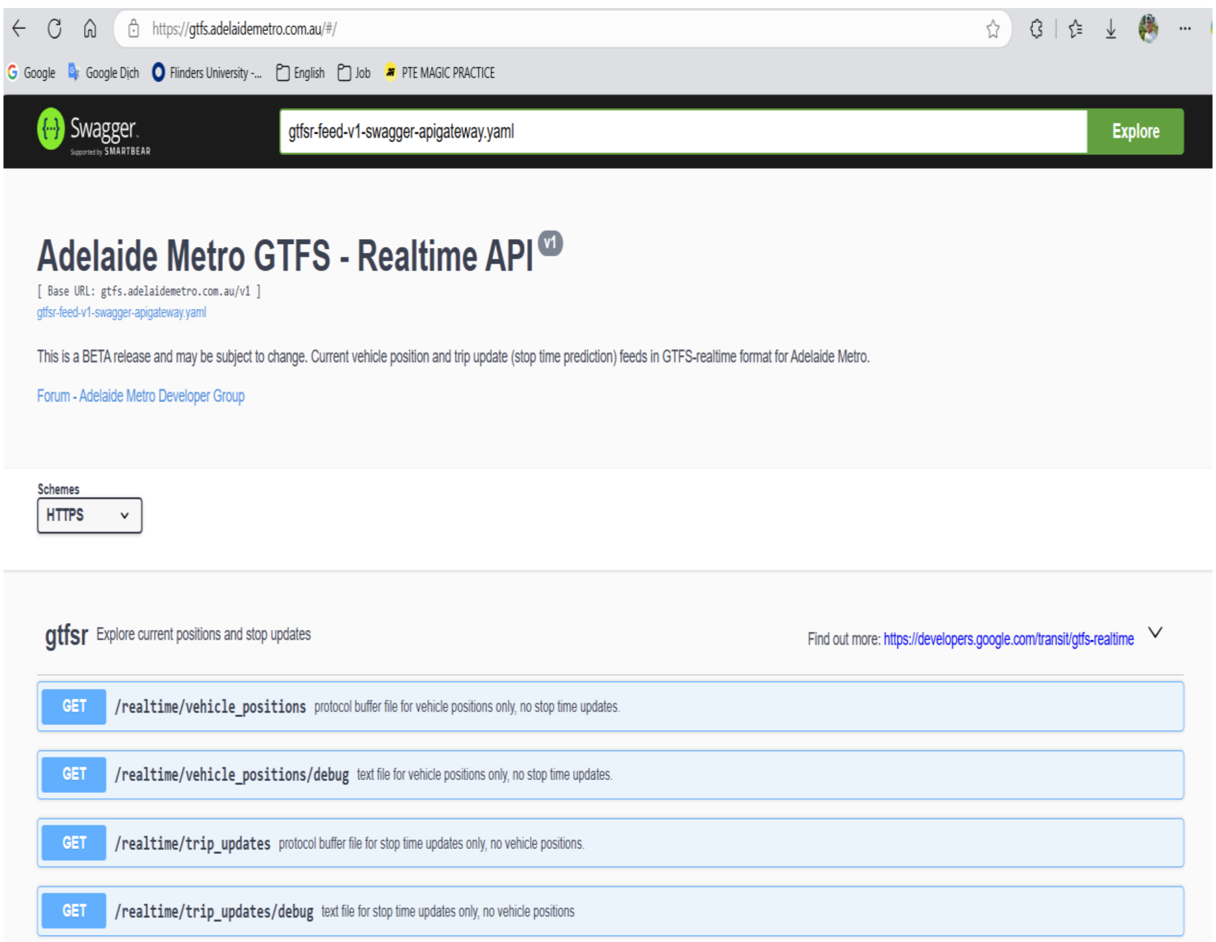

Figure 1: List of API service in Adelaide Metro API

2. Clean data after getting from Adelaide Metro API

Some detected things from the raw Json data from Adelaide Metro API:

- All vehicles all have air conditioned is true and wheelchair accessible is 1 => unnecessary information.
- The "timestamp" was later than current time about 9 hours 30 minutes. That is reasonable because the data centre can be in other country different time zone to Adelaide => need to modify time stamp + a range of time to be correct with current time.

The need for a process in cleaning JSON data, remove redundant information.

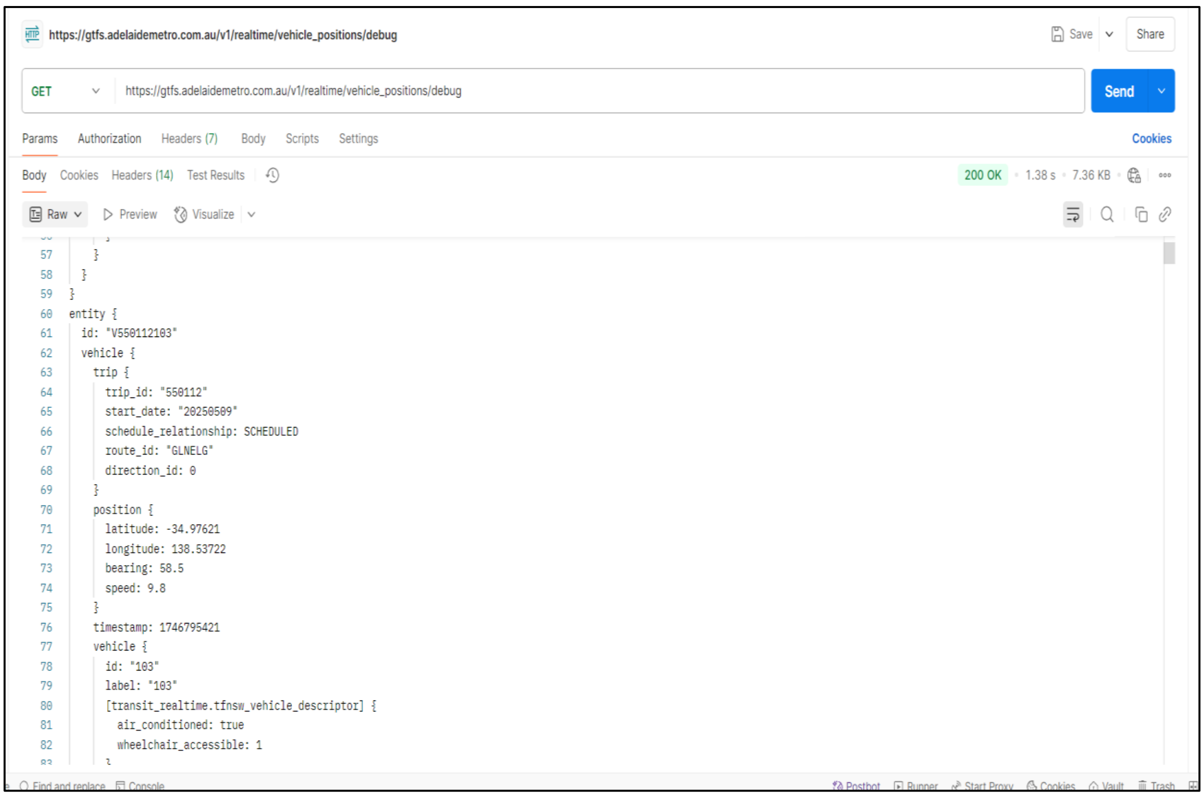

Figure 2: Raw Json data received when calling Metro Adelaide API by Postman

3. Design a web API cloud service as the infrastructure for the system.

From cleaning, retrieving, analysing data, storing it in a database to automation and visualization tasks, there is a need to build a webservices API for the system.

4. **Store processed data in non-SQL database cloud service.**

Based on the data's structure included some fields need to collect such as: id trip, latitude, longitude, speed, route, direction which were not very complex. The need to choose a non-SQL database storage.

5. **Store report files into a file storage cloud service.**

The outcome of analyst's work is reports files such as results files, images as charts, etc. These files need to be stored into storage cloud service to serve many stakeholders such as Adelaide Metro management, Orange's managers, investors.

6. **Monitor system by cloud service**

A cloud service is needed to monitor the system in real time, measure system metrics, and detect attacks.

7. **Build a user-friendly web interface**

Currently, Adelaide Metro's website did not provide website for public user to watch their expected route visualisable. This is also the solution for ordinary users can search for vehicles, view current location and travel history.

8. **Use Node-RED**

Node-RED is the primary tool due to the requirement of Adelaide Metro Management.

Constraints

- Budget constraints: due to characteristics as demo project, team use free or limited versions of Azure cloud services.
- Hardware resource constraints: Node-RED was run in a localhost environment instead of deployed in the cloud for security.
- The vehicle's data collection period was limited to April 2025, so the dashboard only reflects a representative operating cycle.

Team Members and Responsibilities

The Orange's project team consists of 6 members who are talented in various fields

Table 1: Project members and responsibilities

Personnel - position	Responsibility
Le Thanh Nguyen Solution Architect	Design the overall system architecture; main responsibility in the project; setup and deploy resources into cloud services; design Node-Red to store report files to storage cloud service; write code for the API webservice and website.
Nguyen Quyet Thang Mai Data Engineer	Build data pipeline; processing and support the construction of filtering process; design the structure of database; design Node-Red to request database from API and storing into cloud database.
Minh Nguyen Ho Data Analyst	Design algorithms for data analysing, apply analysing skills to find the insight in the whole data; design Node-Red to analyse data from CSV sources file; design Node-Red to analyse data from database cloud service and display to dashboard.
Sarita Ghale System Administrator	Operate system daily, monitor metric of the pipeline system; design security approaches for system; design Node-Red to send report email to users.
Phan Hoai Khoi Nguyen Software Engineer (front-end)	Develop user interface for website and dashboard monitoring, design visualisation interface; design Node-Red to display dashboard for monitoring.
Nachiketkumar Patel Software Engineer (back-end)	Write code for the API webservice; maintain and update system in the backend; design the Node-Red to filter data requested from Adelaide Metro API and connect to other nodes

List of Tools used

Table 2: List of tools used

Tools	Description
Visual Studio	Visual Studio 2022 is a powerful integrated development environment (IDE) (Microsoft, 2022). In project, team used Visual Studio 2022 to develop a backend API to collect raw data from Adelaide Metro API, process and store data in NoSQL database, export data reports in CSV or JSON format.
POSTMAN	Postman is an API development platform that provides tools for efficient API design, testing, documentation, and monitoring (Postman, n.d.). Team used Postman to **test requests to the Adelaide Metro API and test requests to project's web service APIs**. The purpose is to ensure the correctness and performance of the API endpoints.
Azure Data Studio	Azure Data Studio is a cross-platform database development and management tool that supports connections to database management systems such as SQL Server, Azure SQL Database, PostgreSQL, MySQL, and Cosmos DB (Microsoft, n.d.). Data analysts on the team used Azure Data Studio to **query and interact with data stored on Cosmos DB** for data analysis and visualization.
Microsoft Azure STORAGE EXPLORER	Azure Storage Explorer is a application that helps users easily manage data stored on Azure Storage (Microsoft, n.d.). This tool is used by Orange's data analyst and Adelaide Metro's managers to **upload and download report files, as well as other files related to the project**.

Tools	Description
tableau from Salesforce	Tableau is a modern visual analytics platform that helps users discover, manage, and share insights from data quickly and efficiently. Team used Tableau **to create interactive dashboards that display metrics** such as average speed, number of vehicles in operation, stop times at stations
Node-RED	Node-RED is a flow-based visual programming tool that allows users to easily connect hardware devices, APIs, and online services (Node-RED, n.d.). Application in the project: Node-RED is used to interact with the system, including activating data storage functions, sending email reports, monitoring operational results, and managing data files.

Application Overview

The Adelaide Metro Mining Project is built on Microsoft Azure, using cloud services to collect, process and store real-time public transport data. Stakeholders such as system administrators, data analysts and Adelaide Metro managers will interact and operate the system through the Node-RED visualization tool and other supporting tools, to automate monitoring, reporting and data analysis.

Cloud Services Used and Operations

Table 3: List of cloud services used

Cloud Services	Responsible
Azure Functions	**Azure Functions** is a serverless platform that allows to run code that reacts to events without managing server infrastructure. It supports multiple programming languages and integrates easily with other Azure services, Microsoft. (n.d.). Team projects use it for these activities:

Cloud Services	Responsible
	- Process raw data from Adelaide Metro API. - Transform and normalize data before storing. - Create APIs for data retrieval and export.
	Azure Cosmos DB is a globally distributed NoSQL database service that supports multiple data models and provides low latency with flexible scalability. In the scope of project, Cosmos DB is used to store real-time vehicle data, enabling fast queries and supporting effective data analysis.
	Azure Storage provides object, file, disk, queue, and table storage services, which help store data securely. Azure Storage is used to store reports, dashboards, and other data related to data mining and analysis.
Application Insights	**Azure Application Insights** help to monitoring service that provides real-time insights into application (Microsoft, n.d.). Application Insights is integrated to monitor the performance of backend services, helping to detect and fix problems, security.
App Service web app	**Azure App Service** is a fully managed web application hosting that enables to deploy and operate web applications without managing server infrastructure. In the project, this Cloud service is used as hosting for a website where public users can: - Look up real-time vehicle locations for every 15s. - Find routes, travel history and average speeds. - Send feedback or requests to Adelaide Metro management. The website's address was provided as: https://adelaidemetrorealtime.azurewebsites.net/

Business Applicability

The whole system operation is divided into 2 phases when applying into business:

Development Phase

Step 1: Requirements Analysis and Planning

All members in team conduct internal meetings to understand the requirements of the Adelaide Metro client; research data structure from Adelaide Metro's API; evaluate the budget (free/demo service).

Step 2: Design system architecture

Solution Architect designs the overall architecture of the system; select suitable Azure cloud services: Azure Function, Cosmos DB, Storage, App Insights…; assign specific tasks to each technical member.

Step 3: Design Data Pipeline and Processing

Data Engineer designs the data type as JSON and stored in Azure Cosmos DB; build a data processing pipeline, including algorithms for filtering and cleaning raw JSON data from Adelaide Metro API (filtering out unnecessary fields, converting timestamps, etc.).

Step 4: Design logic code and Interface Development

Software Developers (back-end & front-end) write source code to implement APIs using Azure Functions; design processing flows using Node-RED (sending HTTP requests, triggering by hour, sending emails, etc.); build a website that allows users to look up information.

Operation Phase

Step 1: Collect and store data

System Administrator uses Node-RED to trigger the data fetching from Adelaide Metro API. Data is filtered and stored in Cosmos DB according to customer-specific requirements: can get data daily, or filter by route and time.

Step 2: Analyse and extract reports

Data Analyst uses Azure Data Studio, Tableau and supporting tools to retrieve data from Cosmos DB. Perform analysis to provide insights such as:

- High/low traffic routes.
- Peak times, average travel speeds.
- Export reports as CSV or dashboard images, save to Azure Storage.

Step 3: Use results by customers and general users

- **Orange's client as** Adelaide Metro Management will access reports stored on Azure Storage to serve traffic management decisions.
- **General users (passengers)** access the website interface to look up vehicle locations, routes, and send feedback.

Step 4: System monitoring and maintenance

System Administrator monitors system operating status through Azure Application Insights; send reports to Solution Architect when there are problems such as: request errors, data storage failures, or service overload.

The overall architecture is provided below:

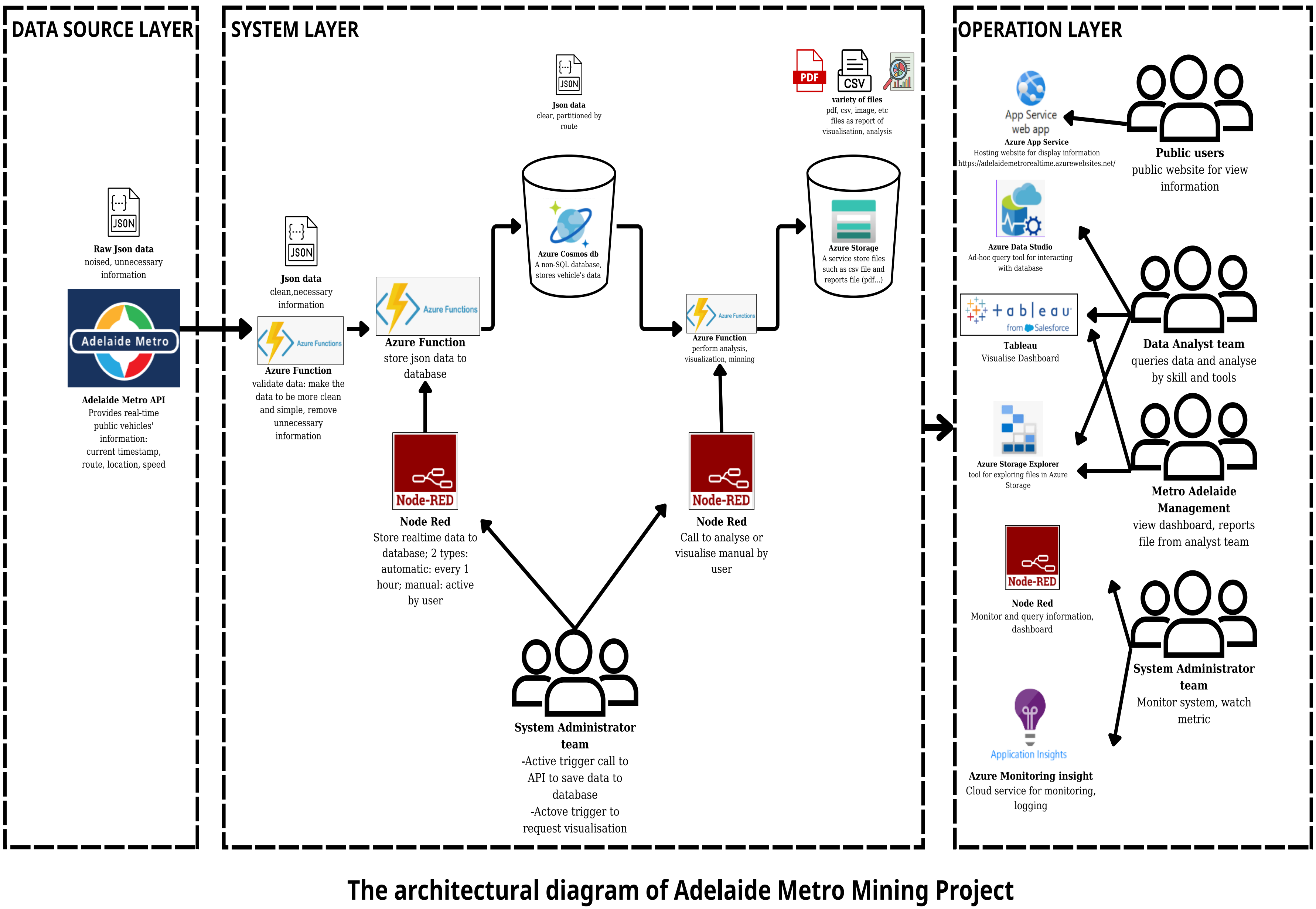

Figure 3: The architectural diagram of Adelaide Metro Mining Project

Practical Functionality

Adelaide Metro Mining Project is designed to provide practical functionality that effectively supports the monitoring and analysis of public transport data. Key features include:

- Use Node-RED to collect data from the Adelaide Metro API, allowing real-time tracking of public transport vehicle locations and status.

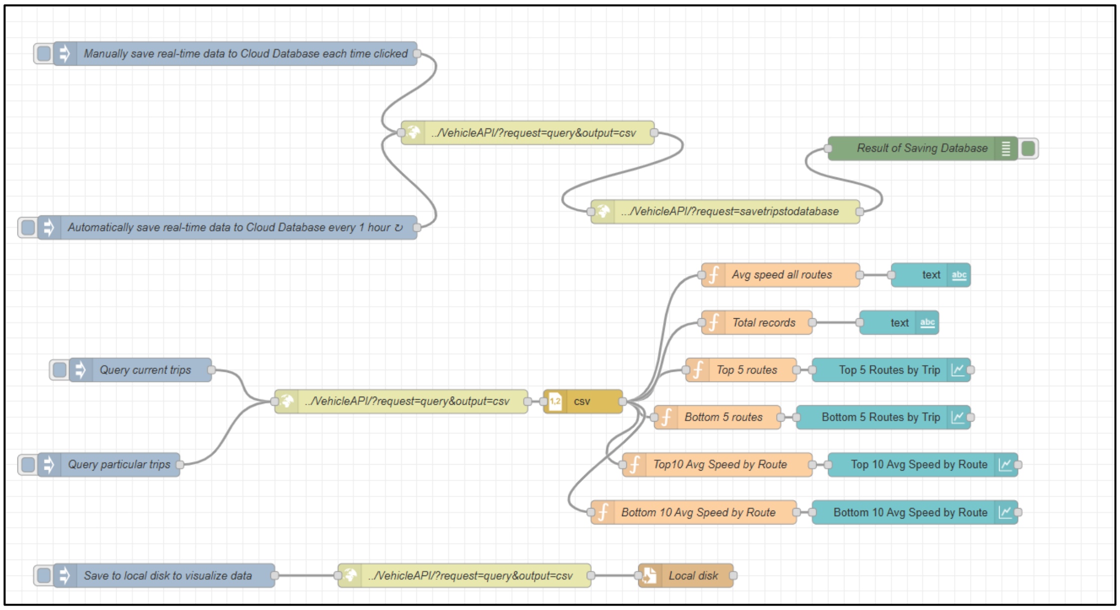

Figure 4: Demonstration of storing, saving, and visualising data in Node-Red

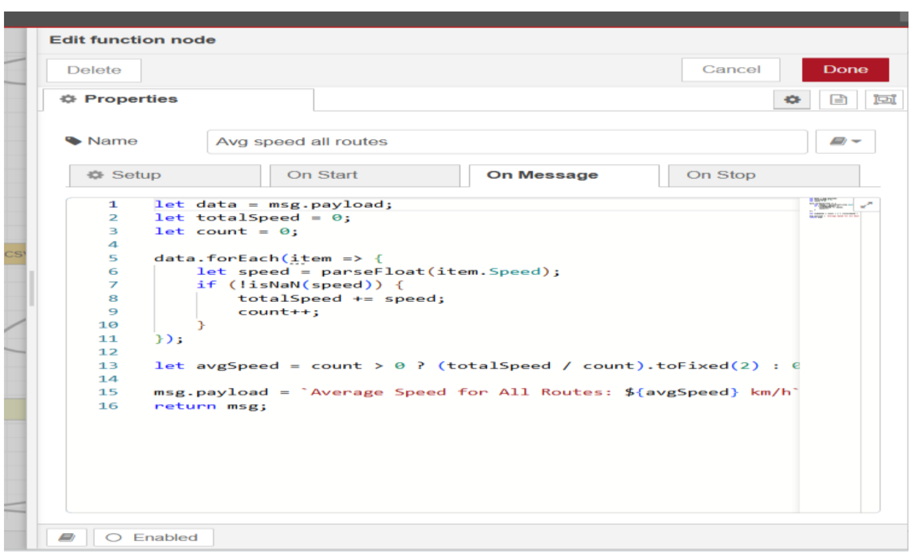

Figure 5: Demonstration of using Function node to filter vehicles have max speed in Node-Red

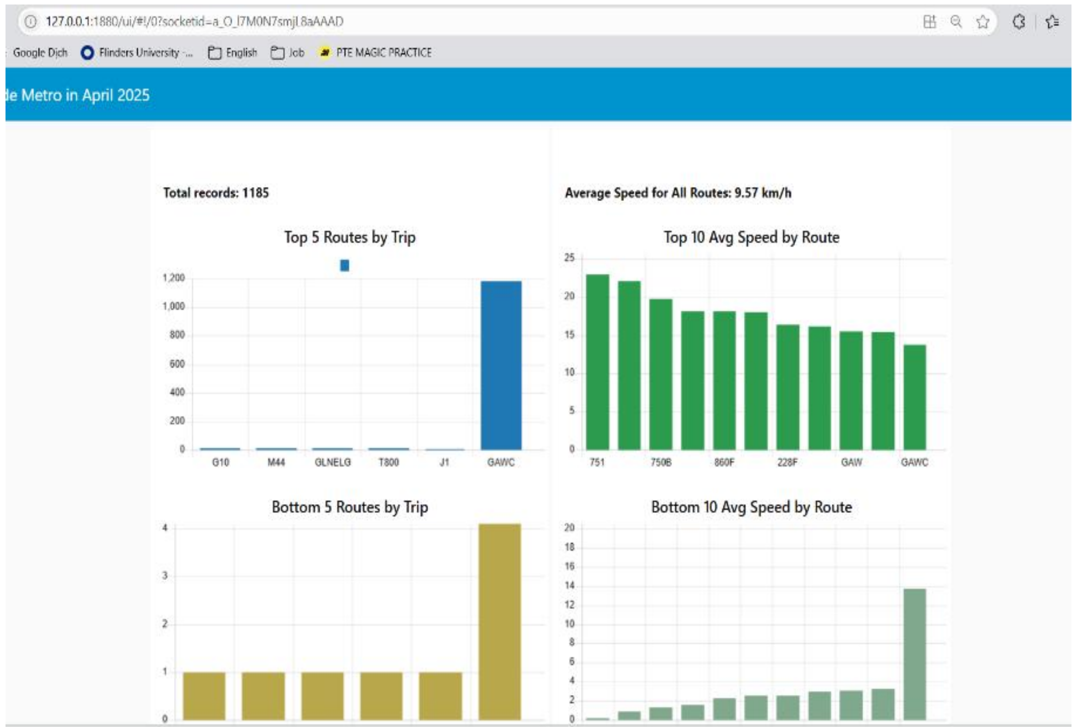

Figure 6: Demonstration of dashboard in Node-Red

- Historical data analysis: Store data in Azure Cosmos DB, allowing querying and analysis of historical data to identify trends and operational performance.

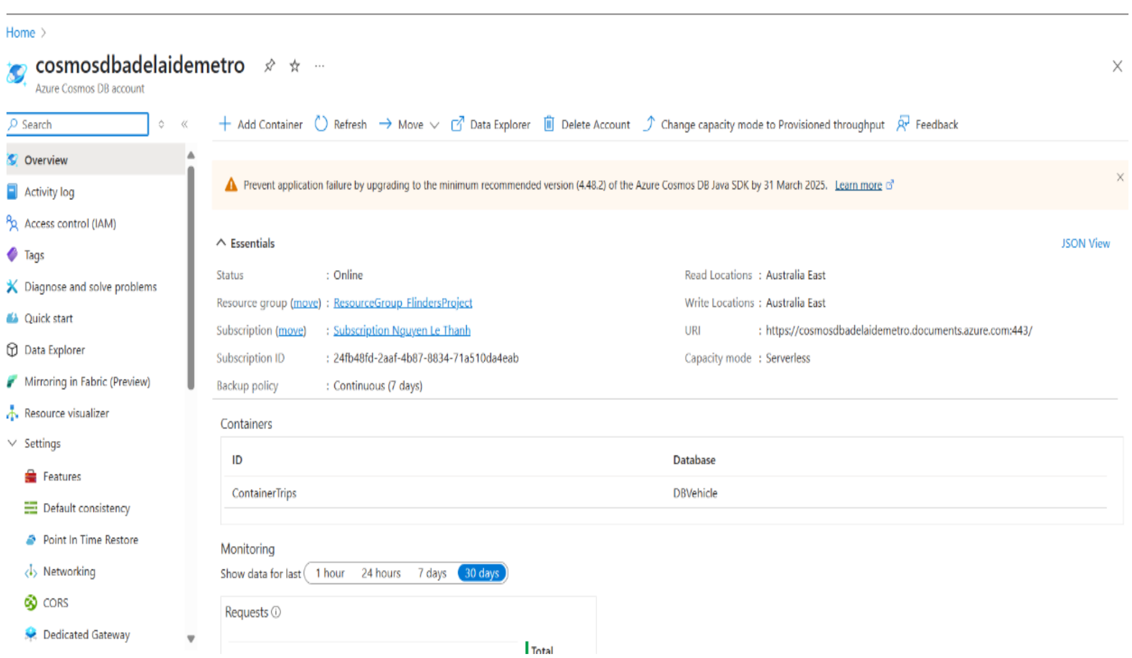

Figure 7: Overview CosmosDB in portal

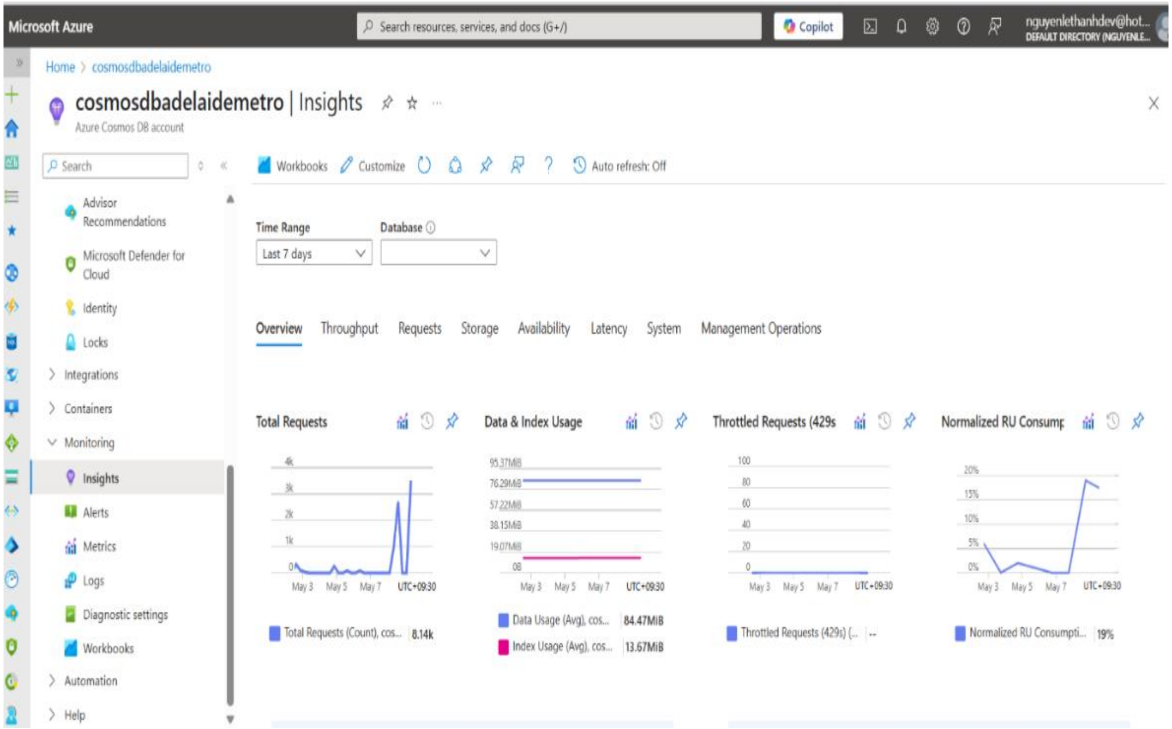

Figure 8: Monitoring total requests and metric data in Cosmos DB portal

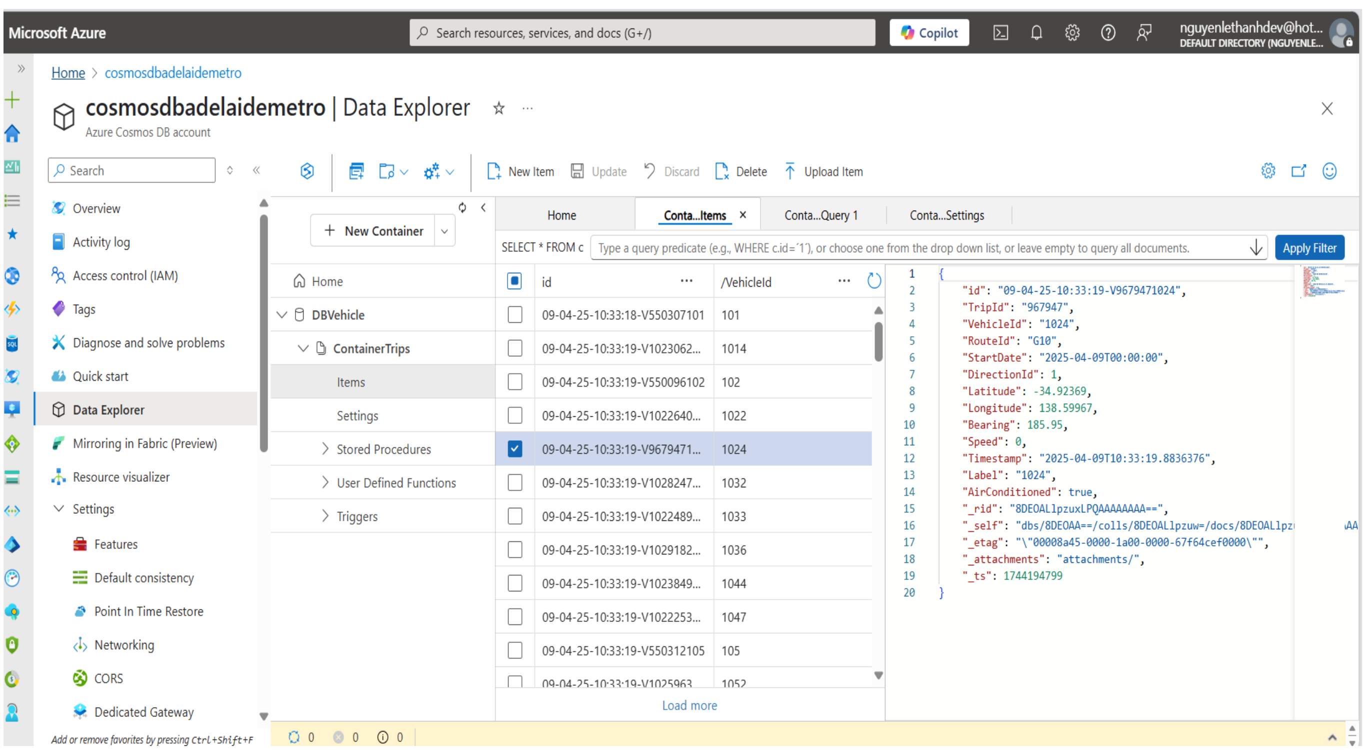

Figure 9: Display data in Cosmos DB portal

- Data visualization: Use Tableau to create interactive dashboards, displaying metrics such as average speed, number of vehicles operating, and congestion points.

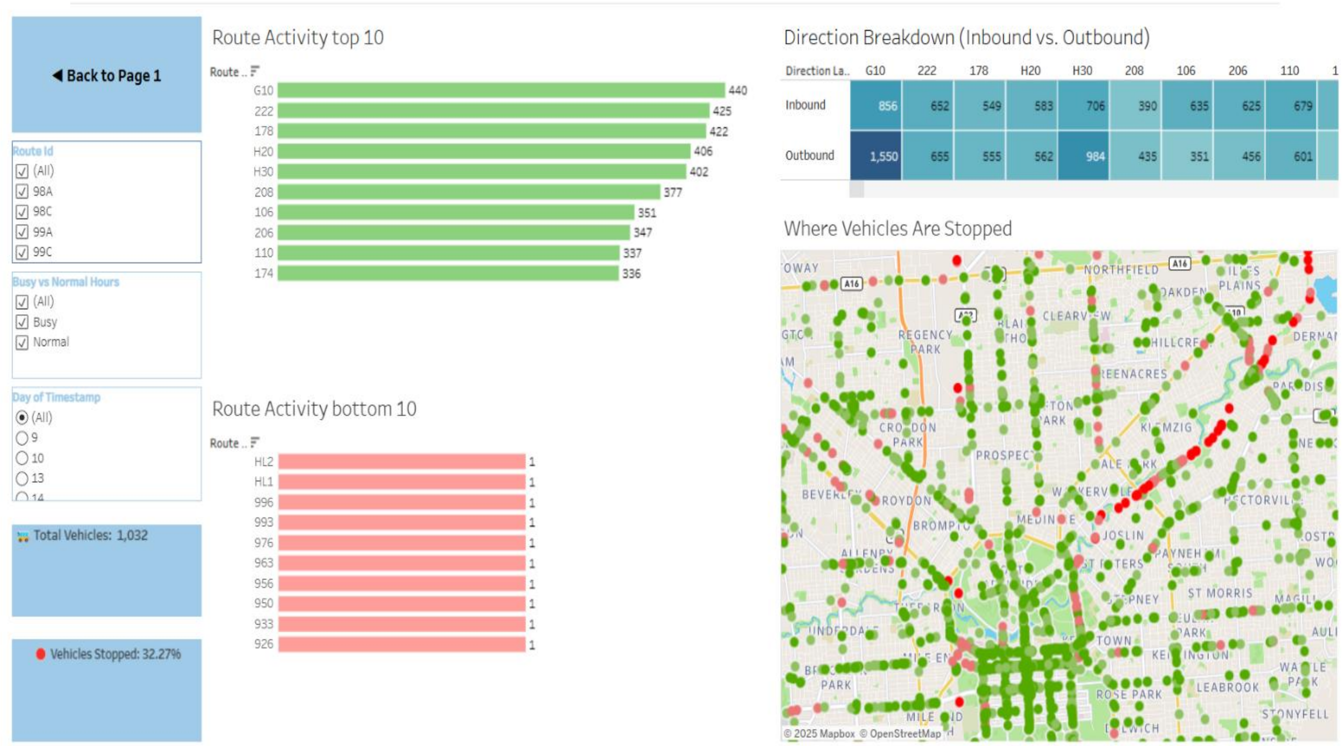

Figure 10: Tableu visualises data from analysing process

- Automatic alerting: Set up alerts in Node-RED to notify when anomalies are detected in the data, supporting timely response.

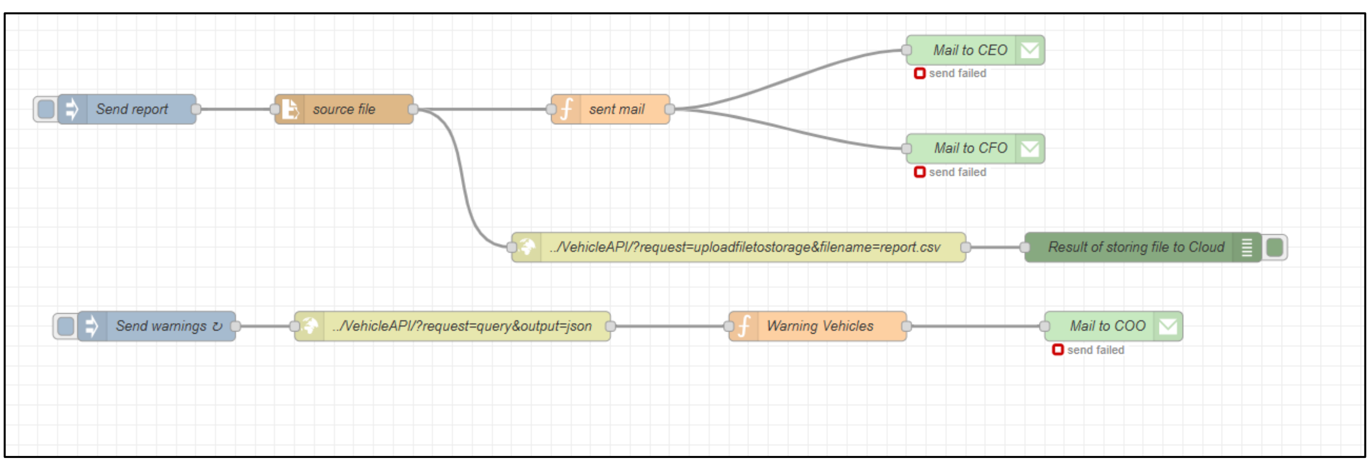

Figure 11: Flow of sending email to manager in Node-Red

- User-friendly interface: Frontend website development allows users to look up vehicle information, routes, and view travel history intuitively.

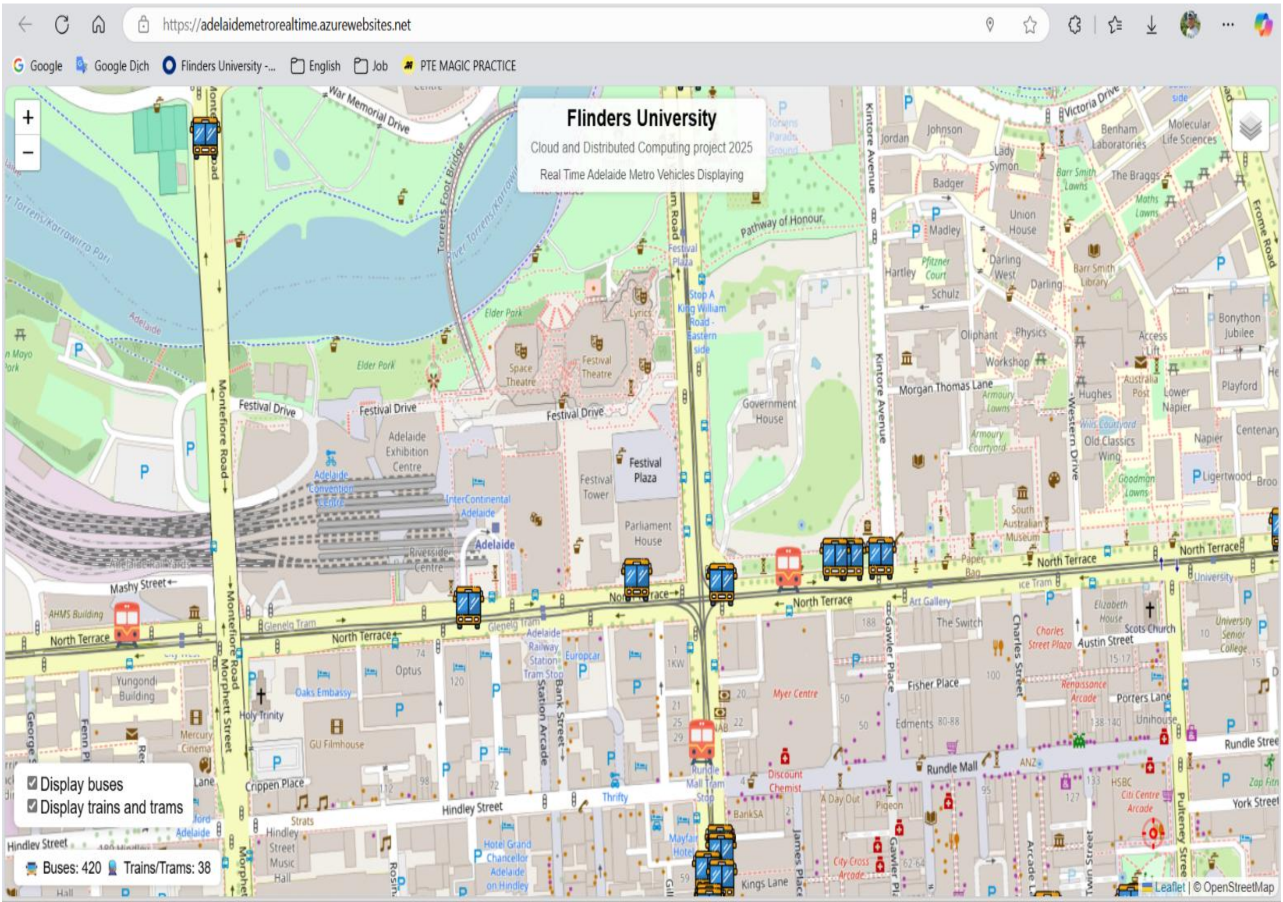

Figure 12: Website for public users to watch real-time vehicles

Value Proposition and Comparison

The Adelaide Metro Mining Project delivers significant value to both transport managers and passengers:

For Adelaide Metro transport managers:

- Provides operational data monitoring and analysis tools to support strategic decision making.
- Reduces incident response time through automated alerting.
- Optimizes resource allocation and operational schedules.

For passengers:

- Provides real-time information on vehicle location and status.
- Improves user experience through an intuitive and easy-to-use interface.

Comparing Adelaide Metro Mining Project and current solution from Adelaide Metro:

Table 4: Adelaide Metro Mining Project compare to current solution in criteria

Criteria	Current solutions	Adelaide Metro Mining Project
Real-time monitoring	Yes	Yes: from Node-Red, Project's website (https://adelaidemetrorealtime.azurewebsites.net), Azure Application Insight
Analyse data history	Limit	Yes: based on CosmosDB database cloud for collecting data)
Visualisation	Yes	Yes: from Node-Red (Dashboard node), Tableau
Automatically alert	No	Yes: based on Node-Red (Email node)
User Interface	Complex	Simple: based on Node-Red, portal website for managing cloud services

Potential Security Vulnerabilities

Potential security vulnerabilities:

- Cyber-attacks: The system can be attacked from outside if not properly protected, especially APIs and web services due to the weak of security in authentication in project's APIs (prototype project).
- Data leakage: Sensitive data such as either personal information in Adelaide Metro or Orange company can be leaked if not encrypted and access rights are not tightly managed.
- Lack of security updates: Failure to regularly update security patches can leave the system vulnerable to attacks.

Prevention solution:

- Use secure protocols such as HTTPS and data encryption.
- Perform regular security checks and update software regularly.
- Tightly manage access rights and use multi-factor authentication for all members (suggest using Microsoft IAM for accessing into cloud services).

CONCLUSION AND REFLECTION

The Adelaide Metro Mining Project was successfully implemented, meeting all technical and business objectives. By integrating Microsoft Azure cloud services such as Azure Functions, Cosmos DB, and Application Insights, along with supporting tools such as Node-RED and Tableau, the system provided real-time monitoring, historical data analysis, and information visualization effectively. Using Azure cloud services helped the team better understand how to build and deploy modern applications, as well as how to optimise system costs and performance.

REFERENCES

- Microsoft Corporation. (2020, September). *Agile Analytics helps Transport for NSW build a better public transport network*. Microsoft Customer Stories. https://www.microsoft.com/en/customers/story/844205-agile-analytics-transport-for-nsw-azure-australia
- Cubic Corporation. (2021). *Real-time passenger information: Umo IQ*. https://www.cubic.com/transportation/products/intelligent-transportation-solutions/real-time-passenger-info
- Node-RED. (2024, June 20). *Version 4.0 Released*. https://nodered.org/blog/2024/06/20/version-4-0-released
- Postman. (n.d.). *Release Notes - Postman*. https://www.postman.com/release-notes/
- Microsoft. (n.d.). *Azure documentation*. Microsoft Learn. https://learn.microsoft.com/en-us/azure/?product=popular
- Tableau. (n.d.). *What is Tableau?*. Retrieved from https://www.tableau.com/why-tableau/what-is-tableau
- IBM. (n.d.). *What is an API?*. Retrieved from https://www.ibm.com/think/topics/api
- Geeks for Geeks. (2025). *What are Web Services?*. Retrieved from https://www.geeksforgeeks.org/what-are-web-services/
- Mozilla Developer Network. (n.d.). *HTTP*. Retrieved from https://developer.mozilla.org/en-US/docs/Web/HTTP
- Wikipedia. (2023). *Cloud computing*. In *Wikipedia, The Free Encyclopedia*. Retrieved from https://en.wikipedia.org/wiki/Cloud_computing
- Two Crows Consulting. (n.d.). *Glossary of data mining terms*. Retrieved from https://twocrows.com/data-mining/dm-glossary/

GLOSSARY

- **Web Service**: is a software system designed to support interaction between electronic devices over a network, using standard protocols such as HTTP to transmit data in the form of XML or JSON (Greek for Geek, 2025).
- **HTTPS** (HyperText Transfer Protocol): is an application layer protocol used to transfer hypertext over the World Wide Web (Mozilla Developer Network, n.d.).
- **API** (Application Programming Interface): An API is a set of protocols and tools that allow software applications to communicate with each other, typically used to access services or data from other systems (IBM, n.d).
- **NoSQL** (Not Only SQL) : is a type of database that does not use the traditional relational model, allowing storage and retrieval of unstructured or semi-structured data, suitable for applications that require flexible expansion and large data processing (MongoDB, n.d.).
- **Serverless**: The cloud computing model allows developers to build and run applications without managing servers; the service provider automatically handles resource provisioning and scaling on demand (Wikipedia, 2023).
- **Data Mining**: The process of discovering hidden patterns, relationships, and trends in large data sets using techniques such as machine learning, statistics, and databases, to make intelligent predictions or decisions (Two Crows Consulting, n.d.).

APPENDIXES

Task 2, 3, and 4 from Week 2 laboratory

Task 2 Search Job in Cloud Computing field

a. **Job search engine used:** https://www.seek.com.au/
b. **Website URL:** https://www.seek.com.au/job/84184507?ref=search-standalone&type=promoted&origin=jobTitle#sol=234cfbcfea8daed2053b1248ad4ef4104449bb11
c. **Post Date:** 13-May-2025
d. **Title:** Solutions Architect - Team Lead
e. **Location:** Adelaide, South Australia
f. **Position Description:** This role requires me to lead a software development team using the .NET platform, design microservices architecture systems and integrate with Azure platforms. The position focuses on developing the company's Employee Relations platform (MyEMS), while coordinating work with offshore development teams and domestic partners.
g. **ICT technologies identified:**

- NET Framework (VB.NET, C#, ASP.NET WebForms)
- Microsoft SQL
- Microsoft Azure (Azure App Service, Azure SQL, Azure Functions, KeyVault, API Management...)
- HTML5, CSS, JavaScript, React
- Agile tools: JIRA, Confluence

h. **Responsibilities/Requirements:**

   i. Installs and maintains: Deploy and maintain responsive web applications using the .NET platform, ensuring performance and quality according to enterprise standards.

ii. Analyses and defines: Analyse and define technical requirements through close interaction with business stakeholders to ensure solutions are aligned with organizational goals.

iii. Proposes and develops: Design and develop microservices, APIs to enhance the flexibility and scalability of the Buzz-ER platform, meeting the changing needs of the business.

iv. Deploys and oversees: Manage the software development process using Agile methodology, coordinate work between offshore development team and domestic development partners to ensure effective system integration.

v. Monitors: Monitor system performance, project progress and source code quality; Proactively propose improvements to software development architecture and processes.

vi. Responsible/Accountable for: Responsible for leading a team of software engineers, providing technical guidance and promoting a culture of continuous learning, while supporting the business technical support team.

vii. Stays informed of: Stays up to date with new technology trends, modern development tools and project management methods such as JIRA, Confluence to improve project efficiency and quality.

Task 3: Banking Services and ICT Technologies

In Australia, banks offer a wide range of digital services supported by modern ICT technologies to improve customer experience and operational efficiency. Three banking services include:

1. **Internet Banking:**

This service allows customers to manage their bank accounts, pay bills, transfer funds, and view transaction history through a secure online portal. It uses web-based interfaces combined with encryption protocols such as SSL/TLS and one-time password (OTP) verification for added security. Users can interact with an portal website to monitor and manage their financial assets.
➡ Example: CommBank NetBank

2. **Mobile Banking:**

Mobile apps offer convenient access to banking services via smartphones or other smart devices such as smart watches. Customers can check balances, transfer money, receive transaction alerts, and even use biometric authentication such as fingerprint or facial recognition. These apps are built on mobile platforms (iOS/Android), utilizing backend APIs and microservices for real-time updates and seamless integration. ➞ Example: NAB Mobile Banking

3. ATM Services:

ATMs enable users to withdraw or deposit cash, print statements, and check balances. These machines are connected to the bank's core banking system via secure LAN or WAN networks and use encrypted communication, chip card readers, and sensors for accurate and safe transactions. However, this service has the potential risk as security due to physical interaction and storing money as notes and coins

➞ Example: ANZ ATM Banking

Benefits:

For customers, these services provide 24/7 access, save time, and offer flexibility in managing finances from anywhere. For banks, ICT integration reduces operational costs, enhances service efficiency, and increases customer satisfaction through secure, user-friendly digital platforms.

Task 4: IBM Cloud

1. **IBM cloud is:**

A cloud platform provided by IBM, including computing, storage, database and development tools. Organizations and programmers can use IBM Cloud to deploy applications without having to manage hardware themselves.

2. **Benefits of cloud computing:**

- Flexible expansion: Easily increase or decrease resources according to demand.
- Cost savings: Only pay for resources used, reducing initial investment costs.
- Access anytime, anywhere: Only need Internet to work.

- Automatic backup and recovery: Data is stored securely, easy to restore in case of problems.

3. **IBM Cloud services support organizations and programmers in useful cloud services:**

- Storage (Cloud Object Storage): Allows storing files, images, videos with large capacity, fast and secure retrieval.
- Code Engine / Cloud Foundry: Helps push code to the runtime environment with just a few commands, automatically manages servers and speeds up expansion.
- Database (Db2, Cloudant, PostgreSQL, ...): Provides database as a service, no installation required, automatic backup, easy to connect via API.

Therefore, organizations save on infrastructure costs, speed up project deployment; programmers only focus on writing code, easily expand and maintain applications.

Task 2, and 3 from Week 3 laboratory

Task 2 Definition of SOAP and REST

SOAP (Simple Object Access Protocol): is an XML-based protocol used to exchange information between computers over the network. SOAP operates based on low-level protocols such as HTTP, SMTP. Communication via SOAP often uses a standard structure, including components such as envelope, header and body. SOAP supports features such as authentication, encryption and complex transactions, so it is often used in large systems or those requiring high security.

REST (Representational State Transfer): is an architectural style used in web services, operating based on HTTP. REST uses methods such as GET, POST, PUT, DELETE to manipulate data. Data is often exchanged in JSON or XML format. REST is simple, lightweight, easy to integrate and is commonly used in current web applications such as Facebook API, Google Maps, and our API in project ...

XML (eXtensible Markup Language): is a markup language used to store and transmit structured data. Data in XML can be easily parsed by both humans and computers. In web services, especially with SOAP, XML is used to define the format of messages sent and received between the client and the server. SOAP messages are often written entirely in XML.

Task 3 Nectar cloud platform

I have familiarised myself with the Nectar cloud platform:

NeCTAR (National eResearch Collaboration Tools and Resources) is Australia's national cloud computing platform, designed specifically for research. Launched in 2012, NeCTAR provides the Australian research community with fast, interactive and self-service access to large-scale computing infrastructure, software and data. NeCTAR provides:

- **Virtual Laboratories**: Enable researchers to access and use research tools remotely, supporting collaboration and resource sharing.
- **eResearch Tools**: Provides software tools to support data analysis and simulation, helping to enhance research efficiency.
- **Research Cloud**: Provides a flexible cloud infrastructure, enabling the efficient deployment and management of research applications.
- **Secure and Robust Hosting Service**: Ensures that research data is stored and accessed securely and reliably.

Task 4 from Week 5 laboratory

IBM Watson is an artificial intelligence (AI) platform developed by IBM, providing services such as natural language processing, text analysis, speech conversion, document understanding, chatbots, etc. Watson helps businesses analyze data and communicate more intelligently with customers.

Watson NLU helps banks analyze text content from sources such as emails, customer feedback, social networks, etc. to identify emotions, intentions, or main topics. For example, an Australian bank can use Watson NLU to automatically assess customer satisfaction from surveys or Facebook comments and quickly detect problems to improve services.

Watson Smart Document Understanding helps banks automatically read and understand content from unstructured documents such as loan contracts, identification documents, invoices, etc. For example: When a customer submits a loan application, Watson can automatically extract information such as name, loan amount, signing date, etc., helping to save time processing documents and reduce manual errors.

IBM Voice Agent helps to build intelligent automated call centers using voice. Voice Agent can listen and respond to customers in natural English. Benefits:

- Reduce the workload for call center staff.
- 24/7 customer support.
- Quick, consistent and accurate responses to common questions such as balance checks, transaction history, product information.